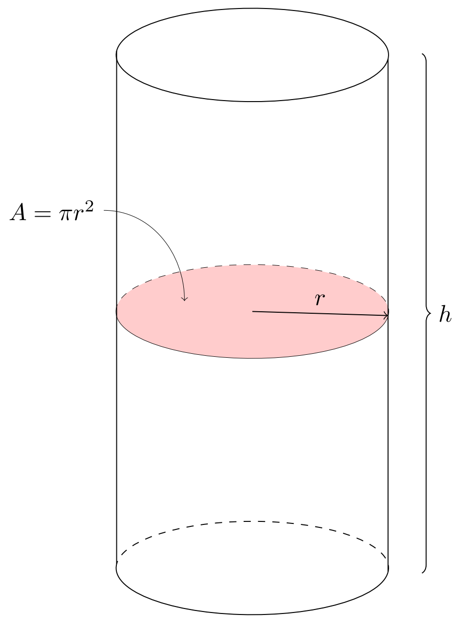
6. Applications of Integrals

6.2. Volumes by Cross-Sections, Disks, and Washers

6.2a Learning Objectives:
- I can think of *cross-sectional area* when calculating the volume of certain shapes using definite integrals.

Recall how we find the volume of a cylinder. $V = \pi r^2 h$. We essentially get the volume by taking the area of the base, πr^2 and sweeping it along the entire height h.

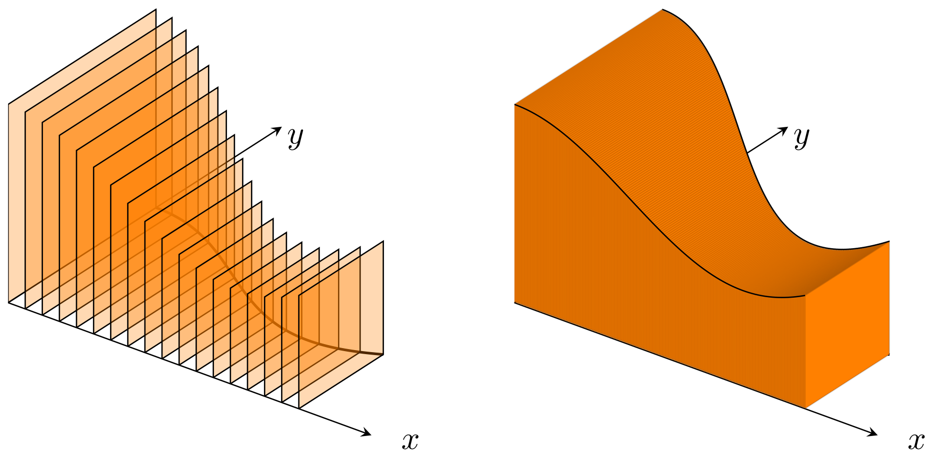


6.2.1. Slicing by Parallel Planes

Definition 6.2.1

We define a *cross-section* of a solid S as the plane region formed by intersecting S with a plane.

We can build a (3 dimensional) solid S by taking the same type of cross-section (e.g. squares) for each value of x, based on some region in the Cartesian plane.

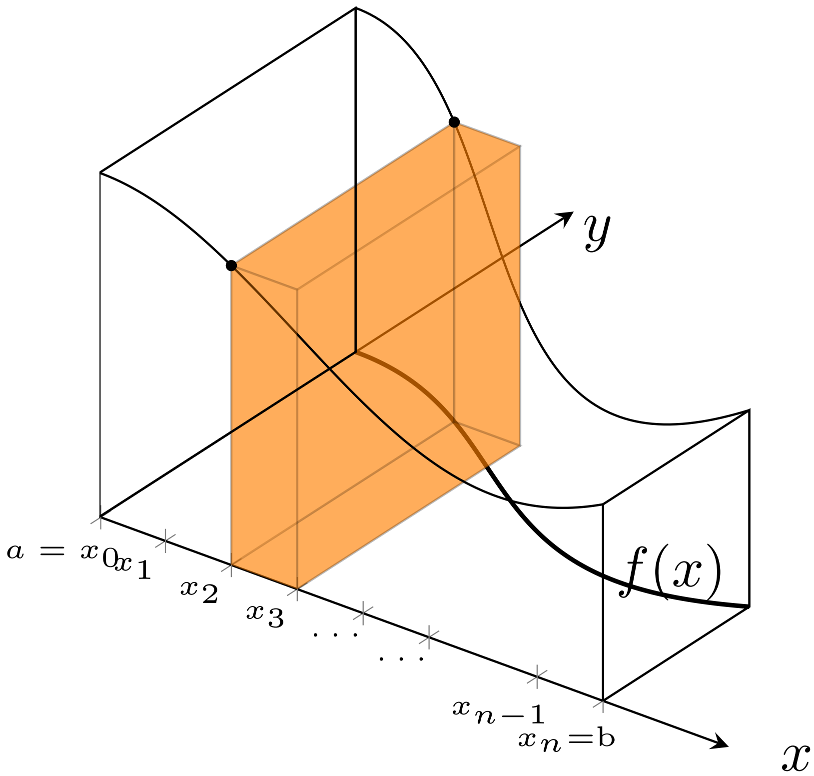


Suppose that we partition $[a, b]$ into subintervals

$$a = x_0 < x_1 < x_2 < x_3 < \ldots < x_n = b$$

where each subinterval has width Δx. What is the volume of one slice of our solid? How would we use this to calculate the voluem of the entire solid?

To approximate the volume, we can think of a cross-section with width Δx, then its volume is $V = (\text{Area of section})\Delta x$. Here the area of the cross-section is the area of a square with side length equal to the function value, $y = f(x)$.



As the width of the sub-intervals goes to zero and we add more rectangular prisms together, we get a better approximation of the volume. Once again, the limit of this sum ends up being a definite integral.

Definition 6.2.2

If the cross-section of a solid S at each point x in $[a, b]$ is a region $S(x)$ of area $A(x)$, where $A(x)$ is a continuous function of x, then

$$V = \int_a^b A(x)\,\mathrm{d}x.$$

Tip

To calculate the volume of a solid

1. Sketch a picture of the region and a typical cross-section.
   - Don't overcomplicate the cross-section, it should just be a prism with straight sides and top/bottom (or left/right sides) the same shape with area $A(x)$ (the area changes for each cross-section but is the same on the 2 faces of the single cross-section). It is never a part of a cone or pyramid or sphere,,,

2. Find a formula for $A(x)$.
3. Find the limits of integration.

- Think about stretching the face with area $A(x)$ from one end of the shape to the other, where do you start/stop this process?

4. Integrate to find the volume, mind your units if applicable.

Note

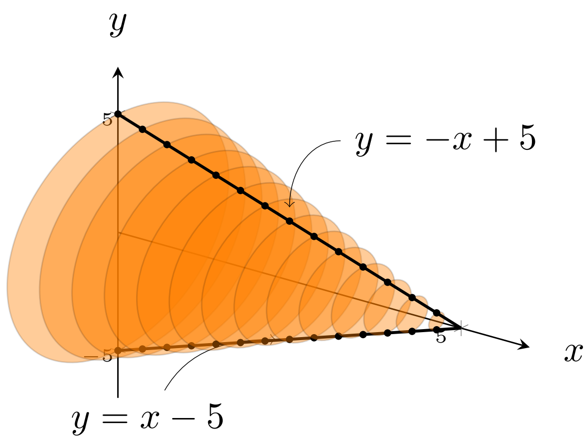
Typical cross-sectional area formulae:
- Square: for a square with side length s, the area is $A = s^2$.
- Triangle: for a triangle with base b and height h, the area is $A = \frac{1}{2}bh$.
- Circle: for a circle with radius r, the area is $A = \pi r^2$.
- Semicircle: for a circle with radius r, the area is $A = \frac{1}{2}\pi r^2$.

Example 6.2.1

A cone with height 5 cm has a circular base with radius 5 cm. Set up a definite integral and evaluate it to find the volume of the cone.

Solution

A sketch of the volume, showing cross-sections, might look like:



We can think of slices of the cone being circles, with radius equal to the y value in $y = -x + 5$. This comes from the edge of the cone making the line $y = -x + 5$ in the x-y plane.

Then the area of one slice is $A(x) = \pi(-x + 5)^2$. The slices range from $x = 0$ to $x = 5$.

The definite integral we need to evaluate is

$$
\begin{aligned}
V &= \int_0^5 \pi(-x+5)^2 \, \mathrm{d}x \\
&= \pi \int_0^5 x^2 - 10x + 25 \, \mathrm{d}x \\
&= \pi \left[\frac{1}{3}x^3 - 5x^2 + 25x\right]_0^5 \\
&= \pi \left[\frac{1}{3}(5)^3 - 5(5)^2 + 25(5)\right] \\
&= \frac{125}{3}\pi
\end{aligned}
$$

We are told the distances are measured in centimeters, so the final answer is the volume is $\frac{125\pi}{3}$ centimeters cubed.

There are 2 checks we can do at the end of the problem. First, our solution is positive, which we expect because volume is positive. Second, this is a cone and we do know the volume formula for a cone:

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(5)^2(5) = \frac{125}{3}\pi.$$

This is the same thing we got from integrating, so our answer is correct!

Exercise 6.2.1

Derive the formula for the volume of a cone using similar techniques as above. *Hint: the line connecting the points* $(0, r)$ *and* $(h, 0)$ *will be useful.*

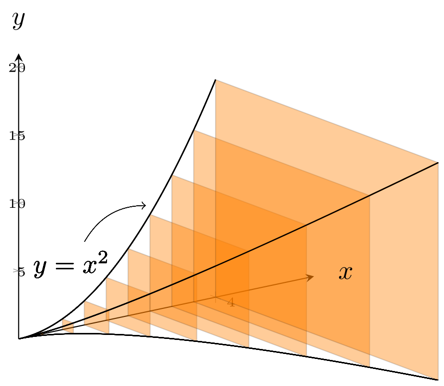
Example 6.2.2

Set up an integral to find the volume of the given solid. The base of the solid is the region bounded by $x = 0, x = 4$, and $y = x^2$. The cross-sections are perpendicular to the x-axis and are squares.

Set up an integral to find the volume of the given solid with the same base and square cross-sections, but the cross-sections are perpendicular to the y-axis.

Solution

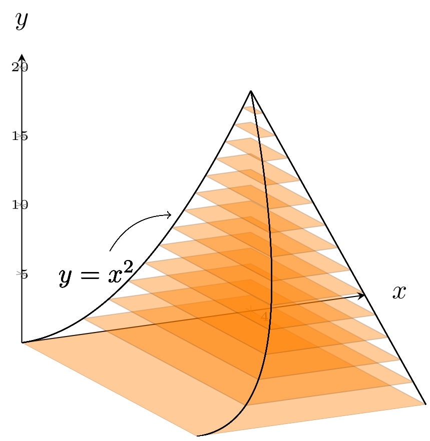
Sketch the region first, then



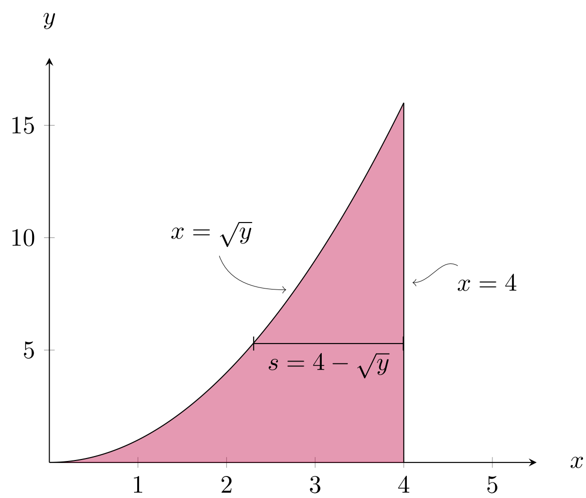
Where we see the side length of the square cross-sections is x^2, and the slices fill the space from $x = 0$ to $x = 4$ so the definite integral we need to evaluate for the volume is

$$V = \int_0^4 \left(x^2\right)^2 \mathrm{d}x = \int_0^4 x^4 \,\mathrm{d}x.$$

For the second part of the problem, we are now looking at the region below:



Notice that the slices now go from bottom to top, when the cross-sections are perpendicular to y-axis, we want to use an integral in y. This region has square cross-sections where we can think about "right curve minus left curve" to find the side length of those squares:



Here the square side stretches between $x = \sqrt{y}$ and the vertical line $x = 4$. So, the side length is $s = 4 - \sqrt{y}$. This solid is formed up of slices from $y = 0$ to $y = 16$. Then the integral we solve for the volume is

$$V = \int_0^{16} (4 - \sqrt{y})^2 \, \mathrm{d}x.$$

Exercise 6.2.2

Predict if you think the volumes of the two solids above will be the same or different. Then, evaluate each integral and compare. Were you correct or incorrect in your prediction? If you were incorrect, that is okay, but try to identify why/what went wrong.

Remark

In these problems, the variable of integration is determined based on whether the cross-sections are perpendicular to the x-axis or y-axis.

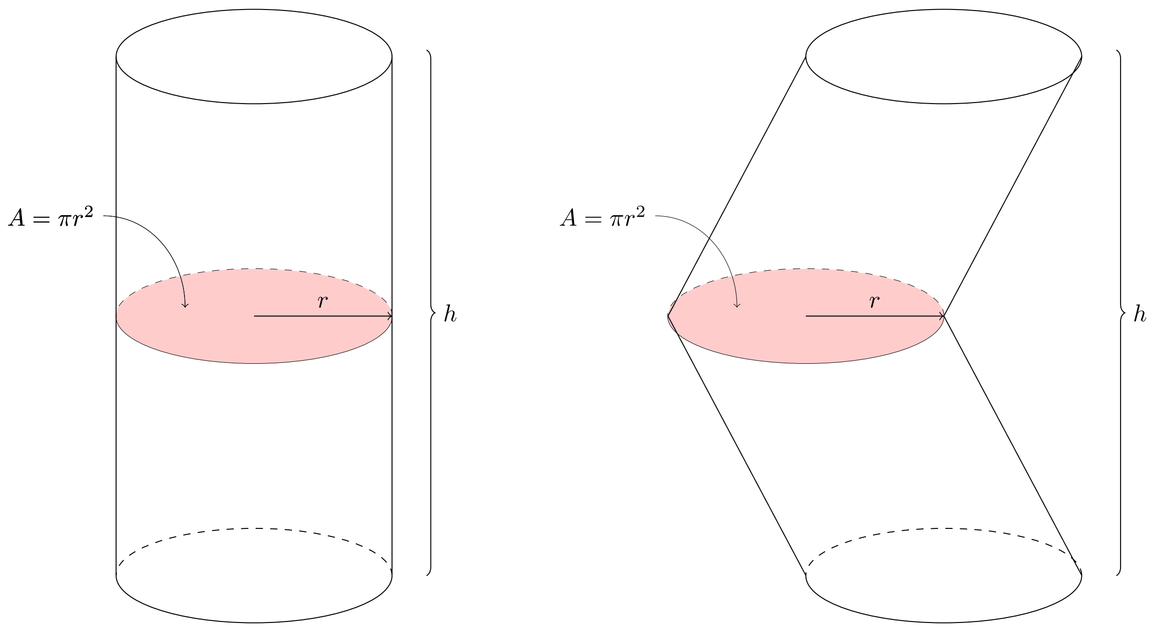
- Slices perpendicular to the x-axis (where slices stack from one x value to another) mean the variable of integration is x.
- Slices perpendicular to the y-axis (where slices stack from one y value to another) mean the variable of integration is y.

Property 6.2.1 Cavalieri's Principle

Solids with equal *height* and identical *cross-sectional area* at each height have the same volume.

Remark

These 2 cylinder-like solids have the same volume by Cavalieri's Principle



Exercise 6.2.3

Set up an integral to find the volume of the given solid. The base of the solid is the region bounded by the graphs of $y = \sqrt{x}$ and $y = \frac{x}{2}$. The cross-sections perpendicular to the x-axis are equilateral triangles.

6.2a Section Summary:

- We found the area of cross-sections and set up integrals to find the volume of solids.